Junho Park

Mail | Homepage | Scholar | LinkedIn | GitHub

Research Interest

My research lies at the intersection of **3D vision, generative AI, and embodied intelligence**. I am interested in developing **foundation models that bridge human-centered perception and robotic manipulation**, enabling robots to learn from human interactions and generalize across objects, tasks, and environments.

Education

Imperial College London, United Kingdom Nov. 2026 – Present
Ph.D., Computing
- Advisor: *Dr. Rolandos Alexandros Potamias*

Sogang University, South Korea Feb. 2024
M.S., Electronic Engineering
- Advisor: *Prof. Suk-Ju Kang*
- Thesis: 3D Hand Dataset Generation Framework with Pose-guided Text-to-Image Diffusion Model [Link]

Sogang University, South Korea Feb. 2022
B.S., Mathematics and Electronic Engineering (Double Major)

Work Experience

LG Electronics, South Korea | *Research Engineer* Mar. 2024 – Oct. 2026
- Collaborated with **Apple** to deploy lightweight vision AI models in **smart home products** on **4-TOPS edge hardware** and develop **automated pipelines for edge-case generation and model training** [Link]
- Developed vision AI models for LG Electronics' **Physical AI initiative** with **NVIDIA**, supporting **large-scale robot data collection and embodied AI development** [Link]
- Contributed to a **multi-task on-device vision foundation model** spanning detection, segmentation, depth, and pose, showcased at **CES 2025** [Link]

Research Experience

VGG, University of Oxford, United Kingdom | *Collaborator* Oct. 2024 – Apr. 2026
- Collaborated with *Dr. Taein Kwon* – Developed a framework for **translating exocentric observations into egocentric views** via multimodal cues (point clouds, 3D hand poses, and language)
- Published 2 first-author papers – **1 ICLR 2026, 1 ICCV 2025 Workshop**

Pusan National University, South Korea | *Collaborator* Jul. 2023 – Feb. 2025
- Collaborated with *Prof. Kyeongbo Kong* – Focused on **3D vision and generative models**, including 3D-aware hand image generation and LLM-based 3D room generation
- Published 4 first-author papers – **1 ECCV 2024 Oral, 1 ECCV 2024 Workshop, 1 ICCV 2023 Workshop, 1 IEEE TMM**

Samsung Electronics, South Korea | *Collaborator* Mar. 2023 – Feb. 2024
- Designed a deep learning pipeline for SEM image restoration and structure prediction
- Published 1 first-author paper – **1 IEEE TIM**

Korea Electronics Technology Institute (KETI), South Korea | *Collaborator* Mar. 2022 – Feb. 2023
- Developed a calibration-free gaze estimation algorithm suitable for public use
- Published 1 first-author paper – **1 IEEE Access**

Sogang University, South Korea | *Master Student* Mar. 2022 – Feb. 2024
- Advised by *Prof. Suk-Ju Kang* – Focused on a wide range of AI, including **3D vision, diffusion models, and LLM**
- Published 8 first-author papers – **1 ECCV 2024 Oral, 2 ECCV 2024 Workshop, 1 ICCV 2025 Workshop, 1 ICCV 2023 Workshop, 1 IEEE TMM, 1 IEEE TIM, 1 IEEE Access**

Research Experience (Continued)

Sogang University, South Korea | *Undergraduate Student* Jul. 2021 – Dec. 2021

- Advised by *Prof. Myoung-Wan Koo* – Developed a speech recognition system for identifying hazardous events in CCTV blind spots, enabling timely response and reinforcing public safety
- Awarded 1st place in **AI Grand Challenge 2021** [Link]

Selected Publications

1. [**ICLR 2026**] **Junho Park**, Andrew Sangwoo Ye, Taein Kwon†. EgoWorld: Translating Exocentric View to Egocentric View using Rich Exocentric Observations. (†: Corresponding Author.) [Project Page] [Paper]
2. [**ICCVW 2025**] Minsuh Song*, **Junho Park***, Suk-Ju Kang†. Replace-in-Ego: Text-Guided Object Replacement in Egocentric Hand-Object Interaction. (*: Equal Contribution, †: Corresponding Author.) [Paper]
3. [**ICCVW 2025**] **Junho Park**, Andrew Sangwoo Ye, Taein Kwon†. Generating Egocentric View from Exocentric View via Multimodal Observations. (†: Corresponding Author.) [Paper]
4. [**ECCV 2024**] **Junho Park***, Kyeongbo Kong*, Suk-Ju Kang†. AttentionHand: Text-driven Controllable Hand Image Generation for 3D Hand Reconstruction in the Wild. (*: Equal Contribution, †: Corresponding Author.) [Project Page] [Paper] **(Oral Presentation, Acceptance Rate: 2.3%)**
5. [**ECCVW 2024**] Jihyun Kim*, **Junho Park***, Kyeongbo Kong*, Suk-Ju Kang†. Interactive 3D Room Generation for Virtual Reality via Compositional Programming. (*: Equal Contribution, †: Corresponding Author.) [Paper] **(Oral Presentation)**
6. [**ECCVW 2024**] **Junho Park***, Yeieun Hwang*, Suk-Ju Kang†. Diffusion-based Interacting Hand Pose Transfer. (*: Equal Contribution, †: Corresponding Author.) [Paper]
7. [**ICCVW 2023**] **Junho Park***, Kyeongbo Kong*, Suk-Ju Kang†. A Novel Framework for Generating In-the-Wild 3D Hand Datasets. (*: Equal Contribution, †: Corresponding Author.) [Paper]
8. [**IEEE TMM**] Jihyun Kim*, **Junho Park***, Kyeongbo Kong*, Suk-Ju Kang†. Programmable-Room: Interactive Textured 3D Room Meshes Generation Empowered by Large Language Models. (*: Equal Contribution, †: Corresponding Author.) [Project Page] [Paper]
9. [**IEEE TIM**] **Junho Park**, Yubin Cho, Yeieun Hwang, Ami Ma, QHwan Kim, Kyu-Baik Chang, Jaehoon Jeong, Suk-Ju Kang†. Mixup-based Neural Network for Image Restoration and Structure Prediction from SEM Images. (†: Corresponding Author.) [Paper]
10. [**IEEE Access**] Joseph Kihoon Kim*, **Junho Park***, Yeon-Kug Moon†, Suk-Ju Kang†. Improving Gaze Tracking in Large Screens with Symmetric Gaze Angle Amplification and Optimization Technique. (*: Equal Contribution, †: Corresponding Author.) [Paper]

Academic Services

Conference Reviewer NeurIPS, ICLR, ICCV, WACV, BMVC

Journal Reviewer IEEE TII, IEEE TCSVT

Honors, Awards, Scholarship

Imperial College London, United Kingdom Doctoral Scholarship Award, 2026

LG Electronics, South Korea Full-funding Scholarship for Master's Program, 2021

Ministry of Science and ICT, South Korea 1st Place, AI Grand Challenge, 2021

Skills

Languages English (fluent), Korean (native)

ML Frameworks PyTorch, Huggingface, PyTorch Lightning, Tensorflow

Data Analytics Numpy, Matplotlib, SciPy, Pandas, Seaborn

SW Engineering Tools Python, C++, Git-based workflow, CUDA, Shell, Linux, Docker, Slurm